Project Initialization and Planning Phase

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to help farmers select the most suitable crop based on soil and environmental parameters using machine learning techniques. The system analyzes agricultural data and provides accurate crop recommendations, helping farmers improve productivity, reduce losses, and increase profits.

Project Overview	
Objective	The primary objective of this project is to develop a machine learning-based crop recommendation system that predicts the best crop for cultivation based on soil parameters and environmental conditions.
Scope	The project focuses on collecting agricultural data, preprocessing it, training machine learning models, and developing a web application that provides crop recommendations to farmers..
Problem Statement	
Description	Many farmers select crops based on traditional methods and experience, which can result in poor yield and financial losses due to unsuitable soil and environmental conditions.
Impact	An incorrect crop selection can reduce agricultural productivity, decrease farmer income, and lead to inefficient utilization of resources such as water and fertilizers.
Proposed Solution	

Approach	Implement machine learning algorithms to analyze soil parameters such as Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall to recommend the most suitable crop.
Key Features	Machine learning-based crop recommendation. Predicts the best crop using soil and environmental data. User-friendly web application for farmers.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5 Processor
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	VS Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset (CSV format))